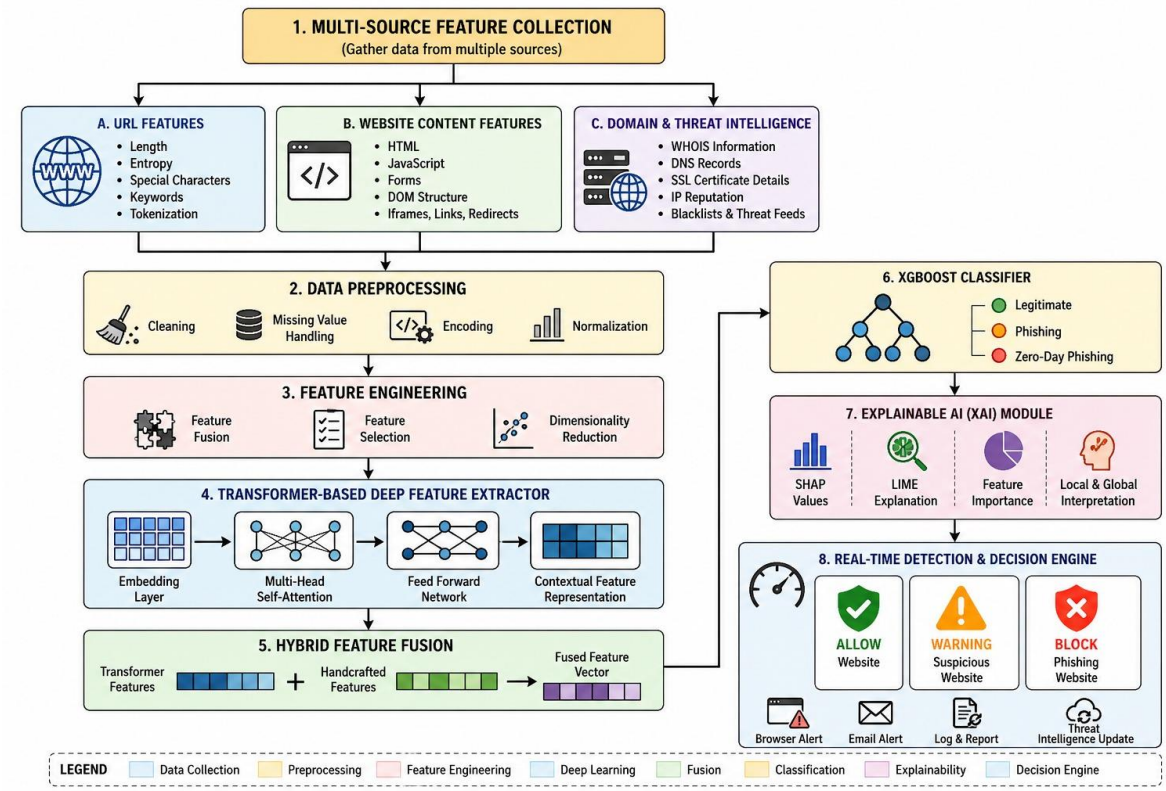
SURE ProEd
CYBER SECURITY & ETHICAL HACKING INTERNSHIP

MINI PROJECT – 3: AI-ASSISTED CYBER SECURITY THREAT MODEL

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1. Multi-Source Feature Collection

The system gathers data about a target website from three distinct angles:

- **URL Features:** Analyzes structural patterns like string length, entropy, special characters, and specific keywords.
- **Website Content Features:** Inspects the raw site data, including HTML, JavaScript, form structures, DOM elements, and redirects.
- **Domain & Threat Intelligence:** Fetches external background data like WHOIS registration details, DNS records, SSL certificates, IP reputation, and existing threat blacklists.

2. & 3. Preprocessing & Feature Engineering

Before feeding this data into a model, it needs to be cleaned up:

- **Data Preprocessing:** Standardizes the data through cleaning, handling missing values, encoding categorical variables, and normalizing numerical scales.
- **Feature Engineering:** Refines the input data by selecting the most relevant characteristics, fusing related attributes, and reducing dimensionality to keep the system fast.

4. Transformer-Based Deep Feature Extractor

This is the deep learning core of the system. It passes data through a **Transformer architecture**:

- An **Embedding Layer** vectorizes the input elements.
- **Multi-Head Self-Attention** and **Feed-Forward Networks** capture deep, context-aware relationships between different website features.
- The output is a highly sophisticated **Contextual Feature Representation**.

5. Hybrid Feature Fusion

The pipeline combines the best of both worlds:

- It takes the deep **Transformer Features** (automatically learned by the AI) and merges them with **Handcrafted Features** (manually defined rules from step 3).
- This results in a single, robust **Fused Feature Vector** containing a comprehensive digital fingerprint of the website.

6. XGBoost Classifier

The fused vector is handed over to an **XGBoost Classifier**, an efficient tree-based algorithm. It categorizes the website into one of three buckets:

- **Legitimate:** Safe to visit.
- **Phishing:** An imitation site designed to steal data.

Zero-Day Phishing: A brand-new, previously unseen phishing threat.

7. Explainable AI (XAI) Module

Instead of acting as a "black box," the system uses **SHAP** and **LIME** frameworks to explain its reasoning. It highlights the exact feature importance and provides local/global interpretations so security analysts can see exactly *why* a site was flagged.

8. Real-Time Detection & Decision Engine

Finally, the system executes an automated response based on the classification:

- **ALLOW Website:** Grants access seamlessly.
- **WARNING Suspicious Website:** Pauses access and flags it to the user.
- **BLOCK Phishing Website:** Immediately drops connection and triggers protective workflows like **Browser/Email Alerts**, logging the incident, and pushing the new signature to global **Threat Intelligence Updates**.

Would you like to dive deeper into any specific section, such as how the **Transformer extractor** works or how **SHAP/LIME** explanations are generated?